# Heatbox — assembly and first run

A hot-air chamber that holds two copper shoes at 110 °C while a third is in use. Printed in Creality Hyper PC.

**Read this before you build it**

This design has never been printed or run. Two of its inputs are assumptions, and both change the part: the heat source is modelled as an adjustable hot-air gun set between 140 and 160 °C. Confirm it, regenerate, then print. The shoes are confirmed as 20 × 40 × 2 mm plates. The temperature gate on the last page is not optional: Hyper PC is rated to about 111 °C and this box runs at 110.

## What you print

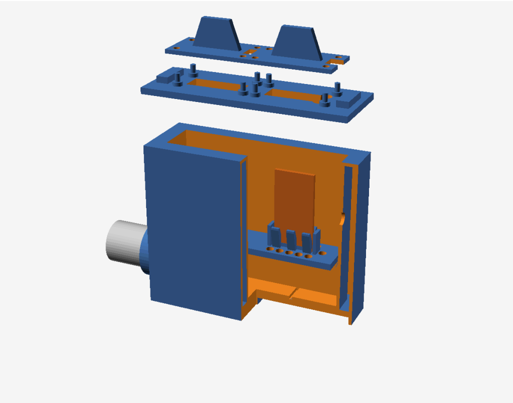
| PIECE | QTY | MASS | SIZE MM |
|---|---|---|---|
| body | 1 | 104 g | 111 × 37 × 85 |
| cap | 2 | 6 g | 42 × 24 × 16 |
| deck | 1 | 7 g | 85 × 19 × 14 |
| lid | 1 | 15 g | 103 × 37 × 12 |
| **total** | | **132 g** | |

Mass is the solid volume of each exported STL at 1.19 g/cm³. Your slicer decides the rest.

## What you supply

| ITEM | WHY |
|---|---|
| Aluminium tube, 20 mm bore × 1 mm wall × 60 mm | The inlet liner. Hot air must land on metal, not on the print. |
| Hot-air source, adjustable 140–160 °C, 20 mm outlet | Drives the chamber. Must sit on its own stand. |
| K-type thermocouple and any reader | Goes in the 6.5 mm port in the right-hand wall. This is the only proof the box is at temperature. |
| Tongs | Shoes are handled hot, never by hand. |

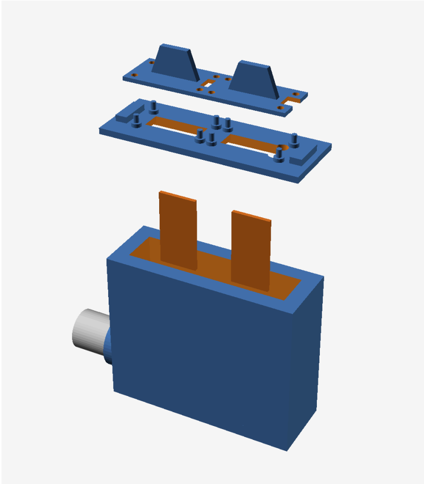
## How it works



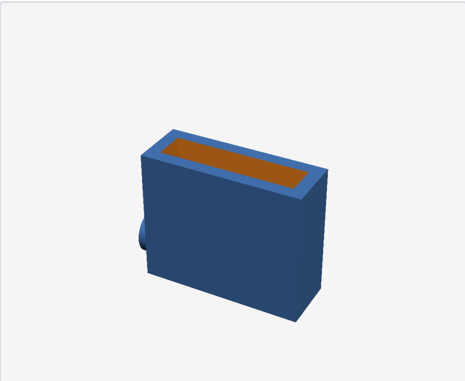
Section through the box, lid lifted off.

Hot air arrives through the aluminium tube into the plenum, the empty space under the perforated deck. It spreads along the bottom, then rises through two rows of 4 mm holes flanking each plate, so the air washes both large faces on the way up instead of hitting one spot. It leaves through the gap under each cap and two vents at the far end. Each plate stands edge-on in a card slot on two low pads, so nearly its whole surface is in moving air — which is why a 2 mm plate recovers in minutes. The wall is two thin skins with a 5 mm sealed air gap between them, and that trapped air is the insulation.

## Assembly

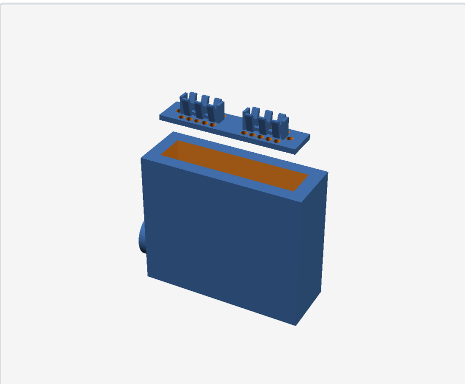


Every piece, in the order it goes together. The grey tube and the copper shoes are not printed.



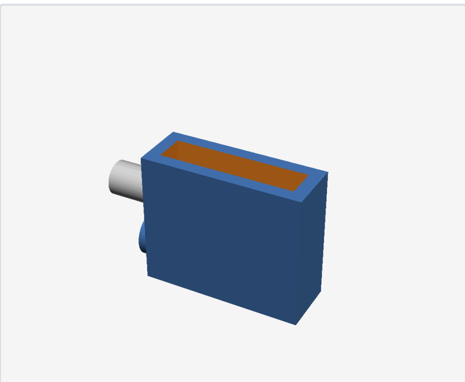
### 1 Print the four pieces

One body, one deck, one lid, two caps. All five sit flat on the bed as exported and none needs support. The body is the long print; the other three are quick. Check the body's underside is a clean closed skirt before you carry on — the floor bridges across it, so a skirt that failed to stick means a sagged floor.



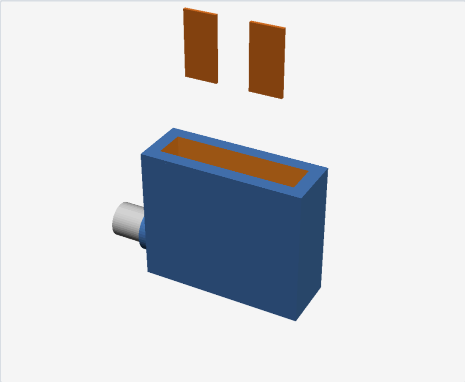
### 2 Drop the deck onto its ledges

The deck falls into the box and lands on six stubs about a third of the way up. It only fits one way round: the end with the single spare hole goes to the right, away from the air inlet. Do not glue it. It lifts out for cleaning, and it is the piece you replace first if anything ever warps.



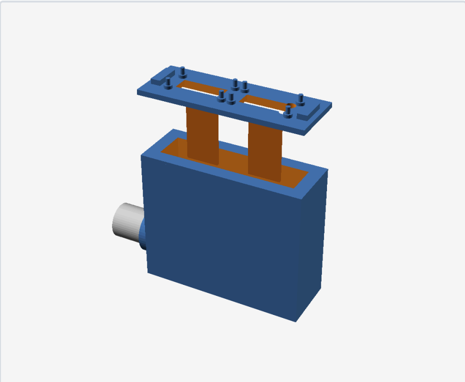
### 3 Fit the aluminium inlet tube

Push the 20 mm bore aluminium tube through the round socket in the left-hand wall until about 25 mm stands inside the box. Three ribs inside the socket hold it; it should be a firm push, not a hammer fit. This tube exists so the hot air from your gun lands on metal instead of plastic, so never run the box without it.



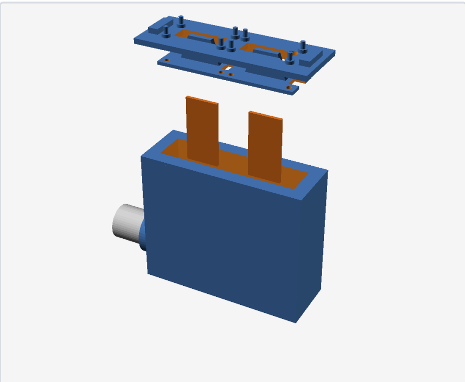
## 4 Slide the two copper plates into their slots

Each 2 mm plate drops edge-first into a card slot: three pairs of short ribs form a groove a little wider than the plate, and it rests on two low pads so its bottom edge stays in moving air. The rib tops flare outward, so you can feel the plate drop in rather than having to aim at the groove — it is well down inside the box. It should seat without forcing. There is a finger of space at both ends of each slot for your tongs, so keep that clear.



## 5 Put the lid on

The lid drops over the rim with its skirt inside the inner wall. Two rectangular slots line up over the two shoes; the two small round vents go to the right-hand end, the opposite end from the inlet.



## 6 Drop a cap over each slot

Each cap sits on four small posts on the lid and is located by four pins. It is meant to sit slightly proud — the 2 mm gap all the way round is the chamber's exhaust, and closing it would pressurise the box. Lift a cap by its fin to take a shoe out; leave the other cap alone.

## First run — do this before you trust it

1. **Probe in, box empty, caps on.** Set the hot-air source to 140 °C and leave it for 30 minutes.
2. **Read the probe.** Between 105 and 115 °C is a pass. Below 105, raise the setpoint towards 160 or increase the airflow. Above 115, lower the setpoint — do not leave it and hope it settles, because the plastic is at its limit there.
3. **Look at the deck.** It is the hottest printed piece and the cheapest to replace, so it is the part that tells you first if the box is too hot. Any softening or sag means stop.
4. **Time a real recovery, once for each pocket.** Take a plate out, let it cool by about 40 °C, put it back, and count the minutes until the probe returns to 105 °C. Expect roughly 2 to 5 minutes. Do this for the left pocket and the right pocket separately: the left one sits over the end of the inlet tube and the probe is in the right-hand wall, so the two do not behave the same. The slower of the two is your swap interval — write it on the box.

## In use

One plate out, two resting. Lift one cap, take that plate with tongs, slide the returning plate into the empty slot, drop the cap back over the pins. The other cap stays shut. Expect a returned plate to need roughly 2 to 5 minutes to come back to temperature — that range is genuine uncertainty about how well the jets reach the plate faces, which is what step 4 of the first run measures and settles.

## Limits — the three ways this breaks

| DO NOT | BECAUSE |
|---|---|
| Rest the weight of the hot-air gun on the inlet socket | Polycarbonate at 110 °C creeps under steady load. Give the gun its own stand. The socket locates the tube, it does not carry anything. |
| Run it without the aluminium tube | Air leaving the source is hotter than the chamber, and it would land straight on plastic that is already at its rated limit. |
| Seal, tape or weigh down the caps | The gap under each cap is the exhaust. Closing it pressurises the box and pushes hot air out through every printed seam. |

## Printing

Creality Hyper PC, nozzle 240–260 °C, bed 50–80 °C, chamber heat on, filament dried first. The model is built around a 0.8 mm nozzle at 0.82 mm line width and 0.4 mm layers: every wall is exactly two extrusion lines thick and every plate a whole number of layers, so the slicer has no leftover slivers to fill. Changing nozzle or layer height means regenerating the STLs with the new numbers, not just re-slicing — the model refuses to build if the walls no longer divide evenly.

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Generated from heatbox.scad by make_guide.py. Diagrams are renders of the same file the STLs come from. Design not yet printed or measured on hardware.